GUQINAGENT: Tool-Interactive Performance-Action Planning for Jianpu-to-Jianzipu Generation

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Abstract

The guqin is a traditional Chinese seven-string zither. Jianpu records pitches and rhythm numerically; jianzipu combines character components to encode performance actions. Jianpu-to-jianzipu generation is thus stateful, constrained performance-action planning rather than direct symbolic transcription. We introduce GUQINAGENT, combining tool-interactive planning, domain-specific fine-tuning, and expert-guided inverse trajectory learning. A teacher uses expert scores to construct executable interactions, redacted into public supervision for single-stage and two-stage agents. The full two-stage system separates basic fingering from idiomatic refinement; both variants use pitch candidates, editable plans, and diagnostic feedback. On 48 scores, the full system achieves 99.981% pitch accuracy within 50 cents on auditor-resolvable events, compared with 86.237% without diagnostic feedback. Our single-stage variant achieves 98.732% and higher reference fingering and ornament agreement. Modern-melody results further show that conditional pitch accuracy and verified-correct output coverage can favor different variants. Evaluation therefore combines pitch fidelity, coverage, reference-action agreement, and runtime diagnostics, with performer assessment planned for feasibility and musical expression. We will release the model, redacted training trajectories, benchmark, and runtime.

1 Introduction

The guqin is a traditional Chinese seven-string zither whose jianzipu notation records performance actions rather than pitches alone. Individual jianzi combine abbreviated Chinese character components to encode string numbers, hui positions along the strings, left- and right-hand fingering, and performance techniques (Li et al., 2026b,a). Jianpu uses numbers and rhythmic marks to specify a melody's pitches and durations. Our input is thus a melody in modern notation, written in numbered notation and expressible in Western staff notation with the same pitches and rhythms.

Realizing a jianpu melody on the guqin requires choosing among alternative string–hui positions and techniques for the same pitch. These choices affect hand configuration, transitions, and timbre; individually correct actions need not form a natural, connected phrase. We therefore frame jianpu-to-jianzipu generation as stateful, constrained performance-action planning rather than direct symbolic transcription. Pitch fidelity and consistency with implemented runtime rules provide checkable evidence, but neither certifies physical playability or idiomatic quality. Slides, vibrato, and compound gestures contribute to *yunwei*, yet more ornaments alone do not imply more expressive playing.

Figure 1 shows five consecutive notes from *Hao Han Ge*. The second note's realization combines a stopped position and an outward pluck with a subsequent left-hand gesture. This single entry specifies both how to sound the note and how to shape it after the attack.

Three challenges follow. *Context-Constrained Performance Planning.* A pitch admits multiple realizations, but selecting one requires jointly considering performance constraints, historical notation state, and phrase context. Harmonic regions and inherited hand positions affect how later entries sound; a locally pitch-correct action need not fit the phrase or preserve subsequent consistency.

Domain-Specific Musical Knowledge Gap. General-purpose LLMs may know technique names and verbal definitions without reliably grounding them in musical meaning, execution, or appropriate use. Such terminology familiarity cannot be assumed to transfer to context-sensitive action selection, motivating domain-specific training rather than reliance on instructions alone.

Missing Process Supervision. Expert jianzipu annotations specify final realizations but generally



Figure 1: Five consecutive notes from *Hao Han Ge* (SHuMJvcc, work unit 7, events 181–185). The left panel pairs input pitch symbols with the generated jianzipu actions in expanded notation. The right panel decomposes the second action into its stopping finger, hui, pluck, string, and subsequent gesture. The illustration was generated from the recorded example.

omit the decisions used to select, verify, and revise actions. Final notation alone does not demonstrate when to query pitch candidates, inspect preceding material, or respond to a warning; these interactions are not directly supervised by final artifacts.

To support context-constrained planning, GUQINAGENTcombines two-stage decomposition with interactive plan editing, following work on reasoning with tools and iterative feedback (Yao et al., 2023; Schick et al., 2023; Madaan et al., 2023). It partitions a complete score into manageable work units while maintaining one shared notation plan. The runtime supplies tuning-aware candidates and conservative checks for pitch, state, and supported performance rules. A Fingering stage establishes a basic realization; a Guqinizer stage considers idiomatic technique and continuity. For domain-specific action learning and process supervision, we fine-tune students on expert-guided inverse trajectories: an offline teacher uses private expert annotations to execute actual tool interactions, which are redacted into student-visible supervision. Automatic checks address verifiable properties, while performer assessment remains necessary for expressive suitability. In summary, our contributions are fourfold:

1. We introduce *constrained jianzipu generation*, a benchmark and executable runtime for reproducible score-to-jianzipu evaluation.

2. We introduce warning-guided inference: deterministic pitch and notation diagnostics are exposed as feedback during generation, enabling the agent to revise its plan.

3. We design GUQINAGENT, with a single-stage trajectory-trained variant and a full two-stage Fingering→Guqinizer agent that distinguishes basic sound production from stylistic refinement. Their comparison measures pitch consistency and reference-action agreement; performer evaluation addresses execution and expression.

4. We develop targeted inverse trajectory construction, which replays annotations through the runtime and exports public, redacted supervision.

2 Related Work

Guqin notation and performance-oriented symbolic music. Guqin notation digitization addresses complementary levels: Li et al. (2026b) recognize the compositional structure of individual jianzi, whereas Li et al. (2026a) extract structured content from full-page documents interleaving jianzipu and Chinese text. These works recover existing notation rather than select new performance actions.

Language-model-based transcription from jianzipu to modern notation includes duration prediction (Shi et al., 2024). Our task is not a symmetric inverse: a melody underdetermines the performance actions needed to realize it.

Guitar tablature inference similarly selects among string–fret realizations of a pitch (Edwards et al., 2024), while piano fingering considers hand placement and sequential execution difficulty (Iman and Ahn, 2024). YuE2 uses editable melody-and-harmony scores to guide audio generation (Yuan et al., 2026). These settings motivate explicit planning, but guqin action selection additionally depends on inherited harmonic regions, hand positions, and compound techniques. Familiarity with technique terminology alone may not support context-sensitive use, motivating guqin-specific training rather than assuming such competence in general-purpose models.

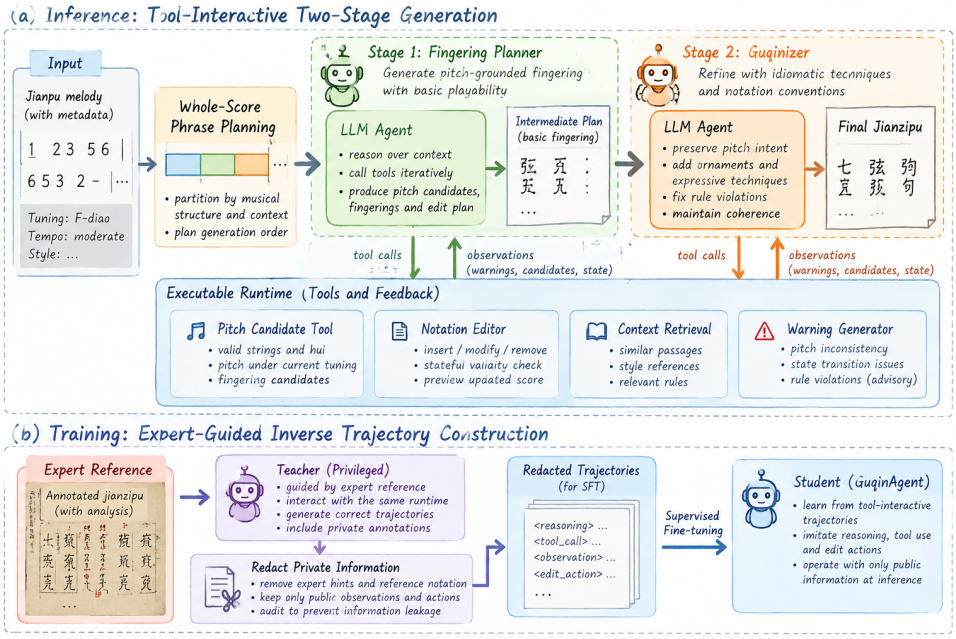


Figure 2: Overview of GUQINAGENT. **Top:** at inference time, the agent partitions a complete jianpu score into ordered work units, constructs a pitch-grounded intermediate plan in the Fingering stage, and refines it into final jianzipu in the Guqinizer stage. Both stages interact with the same executable runtime, which returns pitch candidates, plan previews, compact context, and advisory warnings. **Bottom:** for training, a privileged teacher uses expert annotations only through a private channel while interacting with the public runtime. Redaction retains a student-visible trajectory for supervised fine-tuning, so the deployed student conditions only on score context, its prior actions, and runtime observations.

Tool-interactive planning and revision. ReAct interleaves reasoning and environment actions, while Toolformer and ToolLLM investigate tool-use learning and data construction (Yao et al., 2023; Schick et al., 2023; Qin et al., 2024). Reflexion uses verbal feedback to improve agent behavior across attempts (Shinn et al., 2023). GUQINAGENTmaintains a shared executable notation state and returns event-level diagnostics as the agent edits the plan.

Self-Refine uses model-generated critiques (Madaan et al., 2023), whereas CRITIC grounds revision in external tool feedback (Gou et al., 2024). In GUQINAGENT, the runtime can identify a pitch mismatch without resolving which replacement best fits the surrounding phrase. The agent therefore uses the diagnostic to narrow the edit while retaining responsibility for the performance action.

Learning from interaction trajectories. AgentTuning and FireAct demonstrate the value of supervised interaction trajectories for language agents (Zeng et al., 2024; Chen et al., 2023). More closely related, RetroGen reconstructs process supervision from final expert artifacts for evidence-grounded generation (Huang et al., 2026). Our expert-guided inverse trajectories similarly start from a target artifact, but require the teacher to execute selections, checks, and revisions in the guqin runtime; only public actions and observations enter student training. GEPA instead optimizes prompts by reflecting on execution feedback without updating model weights, providing a distinct comparison to trajectory fine-tuning (Agrawal et al., 2026).

3 Methodology

3.1 Constrained Generation with an Executable Runtime

Given a complete jianpu score $x = (m, e_{1:N})$, the task is to construct an event-indexed jianzipu plan $y = (y_1, \ldots, y_N)$. The metadata m specifies the mode and tuning; each event e_i has a stable index, notated duration, structural type, and a target pitch q_i when defined. An output entry y_i is either a jianzipu expression or an explicit empty display value. The latter may indicate continuation or coverage by a compound gesture and does not remove the event from the score. ABC notation is used only as an internal representation for pitch processing, not as an additional input.

Unlike independent token prediction, the in-

terpretation of y_i depends on preceding actions. We represent this dependence by a deterministic notation-state transition,

$$s_i = \mathcal{T}(s_{i-1}, y_i; m), \qquad s_0 = s_{\text{init}}(m), \quad (1)$$

where s_i records such properties as harmonic-region status, inherited left-hand position, walking gestures, and compound-action coverage. Given this state, the realized pitch $\hat{q}_i = \mathcal{P}(y_i, s_{i-1}; m)$ is a partial function: some compound or ambiguous expressions do not admit a reliable pitch estimate. Structural consistency and pitch agreement are therefore distinct from the broader question of musical suitability.

Interactive plan editing. The complete score is partitioned into an ordered sequence of bounded, musically aligned work units $\mathcal{B}(x) = (p_1, \ldots, p_K)$. This partition is computed from the input score; the benchmark examples remain complete scores. The agent edits one shared plan rather than generating independent phrases. At interaction step t within p_k, its public history consists of the score, current work unit, editable plan, replayed notation state, and previous tool observations:

$$\begin{aligned} h_t &= (x, p_k, y^{(t)}, s^{(t)}, o_{<t}), \\ a_t &\sim \pi_\theta(\cdot \mid h_t, r), \end{aligned} \quad (2)$$

where r denotes the current planning stage and $s^{(t)}$ is obtained by replaying the current plan. The executable runtime $\mathcal{R}$ processes an action and returns the resulting plan, state, and tool observation:

$$(y^{(t+1)}, s^{(t+1)}, o_t) = \mathcal{R}(x, y^{(t)}, s^{(t)}, a_t). \quad (3)$$

Read-only actions leave the plan unchanged. The runtime used to construct training trajectories is also used at inference time.

The agent can request tuning-aware string–hui realizations through `get_pitch_candidates`, including open strings, stopped notes, harmonics, and supported inherited or compound gestures. Returned candidates distinguish exact, approximate, and unavailable matches. The only plan-writing operation is `edit_plan(jianzi_rows)`, which applies event-indexed updates and replays the resulting notation state. Compact context retrieval through `list_context` and targeted earlier-phrase lookup provides prior material without reproducing the entire score in every interaction.

Conservative constraint feedback. We distinguish runtime-checkable violations from cases whose interpretation is uncertain. For a rule c and event i, let $v_c(i; x, y, s) \in \{0, 1, \perp\}$ denote no detected violation, a detected violation, or an indeterminate check, respectively. The warning set is

$$W(x, y) = \{(i, c) : v_c(i; x, y, s) = 1\}, \quad (4)$$

with s obtained by replaying y. In the implementation, checks cover reliably parsed pitch mismatches and conservative diagnostics for harmonic-state continuity, walk direction or span, interrupted open-string resonance, paired stroke adjacency, and difficult left-hand reaches. Indeterminate cases are not converted into violations. Warnings accompany the relevant `edit_plan` preview but neither rewrite the plan nor imply that an unflagged output is musically correct. The agent may query candidates, revise the affected entries, or retain its choice; identical repeated edits are stopped to prevent unproductive loops.

3.2 Two-Stage Performance-Action Planning

The same notation event jointly specifies how a pitch is produced and how it is articulated. We separate these decisions into two successive planning stages. *Fingering* first constructs a basic pitch-grounded realization, choosing strings, hui positions, left-hand contact, and elementary right-hand strokes. *Guqinizer* then revises the committed plan for idiomatic execution, including ornaments, walking gestures, harmonic transitions, and compound-action notation. The second stage may preserve or revise earlier decisions; it is not a post-processing formatter.

For work unit p_k, let $y^{(k-1)}$ be the shared plan after completing the preceding units. A bounded runtime rollout under each stage gives

$$\begin{aligned} y^{F,(k)} &= \text{Rollout}_F(x, p_k, y^{(k-1)}; \mathcal{R}), \\ y^{(k)} &= \text{Rollout}_G(x, p_k, y^{F,(k)}; \mathcal{R}), \end{aligned} \quad (5)$$

where each rollout consists of stage-conditioned tool interactions of the form in Eq. (3). Only edits committed through the runtime enter the shared plan. Crucially, the state passed to p_{k+1} is replayed from $y^{(k)}$, including any earlier mistakes, rather than substituted from an expert reference. Both stages retain access to pitch candidates and runtime feedback. Warnings can trigger another revision

but are not hard acceptance criteria; execution remains subject to the configured interaction budget.

This decomposition imposes a procedural bias: establish a basic sound-producing realization before making the more context-dependent choices involved in idiomatic execution. It does not assume a unique separation between fingering and ornamentation, nor does it guarantee that the second stage preserves every correct decision. We evaluate both the benefits and regressions introduced by refinement against a single-stage agent using the same public tools.

3.3 Expert-Guided Inverse Trajectory Learning

Expert jianzipu annotations specify desired performance actions but generally not the tool interactions by which an agent could arrive at them. Directly supervising final notation therefore leaves the interaction policy unspecified. We instead construct trajectories by allowing an offline teacher to work backward from the annotated target while acting through the same executable runtime available to the student.

Let z be a phrase-level expert annotation and $\mathcal{K}(z)$ the technique-relevant descriptions retrieved from a curated knowledge base. Both are private to the teacher. Its rollout is conditioned on the input, private guidance, and public runtime:

$$\tau^T \sim \mathrm{Rollout}\big(\pi_T \mid x, z, \mathcal{K}(z); \mathcal{R}\big), \qquad (6)$$

where τ^T records the chronological sequence of tool actions, returned observations, and accepted plan updates. The reference guides the search for an executable realization but does not constitute a tool response. The teacher constructs stage-specific traces: basic fingering first, followed by idiomatic refinement. Semantic descriptions clarify technique constraints that may be implicit in compact jianzipu symbols; they are not added to the student's inference context.

Student supervision. We remove private expert guidance from the teacher traces and fit a stage-conditioned student policy to the retained public interaction sequences $\mathcal{D}$. For a supervised token position u in a trajectory $\tilde{\tau} \in \mathcal{D}$, let h_u^{pub} contain the preceding student-visible context and let w_u be its target token. The training loss is standard masked cross-entropy,

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = -\sum_{\tilde{\tau}\in\mathcal{D}} \sum_{u\in\mathcal{Y}(\tilde{\tau})} \log p_\theta(w_u \mid h_u^{\mathrm{pub}}), \quad (7)$$

where $\mathcal{Y}(\tilde{\tau})$ indexes supervised assistant tokens; tool outputs form context but are not prediction targets. At deployment, neither expert annotations nor private semantic guidance are provided. The resulting student must recover its actions from the input score, its own previous edits, and the executable feedback it has observed.

4 Experiments

4.1 Experimental Setup

Data and models. We train on 153 scores (3,976 work units) and evaluate on 48 traditional scores (1,031 units) and 10 modern melodies (91 units). Splits are defined at score level. Each method carries its own generated history across work units, and reference jianzipu is reserved for offline scoring. All students use Qwen3.5-9B with 4-bit LoRA. The two-stage adapter learns jointly from 7,952 stage-specific conversations. Each single-stage recipe uses 3,976 examples. Appendix A specifies training and inference budgets.

Comparisons. We compare with the original Qwen checkpoint in the two-stage agent, GLM-5.3 ReAct and GEPA with the same technique handbook, and Direct SFT. GEPA results are pending (Agrawal et al., 2026). Direct SFT learns the same final teacher plans as our single-stage student and generates once per unit without tools or repair. We also compare fixed-workflow supervision and disable diagnostic feedback in the two-stage system.

Inference protocol. Tool-based Qwen runs permit up to three attempts, with 16 rounds per single-stage attempt or eight rounds per stage attempt in the two-stage system. Feedback ON allows up to three Guqinizer passes, while OFF uses one. GLM ReAct allows eight rounds per attempt and two attempts. These ceilings define the deployed pipelines and can yield different total interaction budgets.

Metrics. PitchAcc@50c measures pitch fidelity on auditor-resolvable events. We also compare methods on shared resolvable events. Position F1 compares exact string–hui pairs, and ornament F1 compares ornament labels, while Tech. Sim. and Tone Sim. compare distributions. Diversity-adjusted Nonbasic Rate discounts the rate of rule-compliant nonbasic technique use by within-score technique repetition (Appendix B). Rule Violation Rate counts audited units with at least one final

violation. Valid-output coverage uses all test units as its denominator. Musical metrics for incomplete runs apply to valid contiguous prefixes. Appendix B specifies the scoring rules.

4.2 Main Results

On traditional scores, both GUQINAGENTvariants complete all 1,031 work units. The two-stage system achieves 99.981% conditional pitch accuracy and a 0.194% rule violation rate, compared with 99.760% and 5.044% for GLM-5.3 ReAct with the handbook (Table 1). Both variants also achieve higher Position F1 and ornament F1 than ReAct on these scores, while ReAct more closely matches the reference tone distribution. Rule violations matter because a pitch-correct plan can still disrupt how a phrase is played or sounds. For example, stopping an open string too soon can cut off its resonance, a slide with the wrong direction can change the intended contour, and a difficult same-finger shift can interrupt continuity. The violation rate therefore captures departures from encoded guqin playing practices that pitch accuracy and reference-action agreement alone can miss.

Direct SFT yields 85.277% pitch accuracy on its valid traditional prefixes, with stronger ornament agreement than either GUQINAGENTvariant. On modern melodies, where Direct has full protocol coverage, its pitch accuracy and rule consistency still trail both variants. The comparison spans both supervision and tool access. The original Qwen agent produces valid outputs for 11 of 1,031 traditional units. Appendix F reports its failure diagnostics, and Appendix D details Direct's coverage and shared-event comparisons.

The baselines also expose a distinction between reproducing reference gestures and realizing the melody. On modern melodies, Direct SFT nearly matches the single-stage student's ornament F1 (43.437 vs. 43.445), yet on the 1,315 events resolved for both, its pitch accuracy is 80.760% versus 98.023%. This gap persists on a common event set even though the methods differ in which outputs the auditor can resolve.

4.3 Ablation Studies

What does expert guidance add to a fixed workflow? Both single-stage students can query pitch candidates and edit the plan at inference. Their training differs in what those interactions demonstrate. Fixed-workflow trajectories contain one scripted query followed by an edit derived from the original annotation. Expert-guided trajectories demonstrate how to select, check, and revise an executable plan. On 12,792 shared traditional events, pitch accuracy rises from 26.931% to 98.781%, even though the fixed-workflow model has higher ornament F1. The gap persists on the modern set (19.929% vs. 98.749% on 1,119 shared events). These comparisons support the expert-guided recipe as a whole. Its trajectories, final targets, and some prompts differ from the scripted recipe.

What does two-stage planning trade off? The two-stage system improves pitch accuracy on shared traditional events and leaves violations in only 2 of 1,031 units, compared with 147 for single-stage (Table 2). The single-stage student more closely reproduces reference actions, including string–hui positions and ornaments. One illustration of why pitch accuracy leaves room for different performance choices is tone type. Among recognized tone types on reference-labeled events, harmonics make up 7.51% of the full system's predictions versus 29.34% in the reference. That difference matters when interpreting near-perfect pitch accuracy. The two variants use separately trained adapters and different interaction budgets, so the comparison evaluates the complete pipelines.

What does diagnostic feedback contribute? Holding the two-stage adapter fixed, enabling diagnostics and their follow-up passes improves accuracy by 13.038 percentage points on 14,289 shared traditional events. The modern set shows the same pattern (98.899% vs. 73.879% on 1,271 shared events). Both settings retain pitch queries and structural checks, and the same final rules score their outputs. On the traditional set, final warnings fall from 3,436 to 2. One residual case in *Youlan* realizes E6 for a target E7, illustrating that a bounded revision loop can still leave a pitch error. This comparison estimates the contribution of the feedback mechanism, including the extra revision opportunities it enables. Appendix F reports full counts and reference-agreement diagnostics.

4.4 Generalization and Efficiency

All six completed methods cover all 91 modern work units. The two-stage pipeline is more accurate on its resolvable events (98.951% vs. 97.932%) and on the 1,248 events both variants resolve (99.920% vs. 98.157%). Single-stage resolves 1,402 of the 1,528 target events, compared with

Method	Pitch Acc. ↑	Position F1 ↑	Tech. Sim. ↑	Orn. F1 ↑	Adj. Nonbasic Rate ↑	Tone Sim. ↑	Rule Violation Rate ↓
(a) Traditional scores: 48 scores, 1,031 work units							
Agent baselines							
Original Qwen3.5-9B, two-stage agent[†]	N/A	N/A	N/A	N/A	N/A	N/A	N/A
GLM-5.3 ReAct	99.760	30.927	0.6323	7.158	8.97	**0.8439**	5.044
GLM-5.3 + GEPA	–	–	–	–	–	–	–
Direct supervision baseline							
Direct SFT[‡]	85.277	37.369	0.7536	27.604	14.79	0.8647	62.612
Ablations							
Fixed-workflow SFT, single-stage	26.778	11.002	**0.8401**	**27.928**	10.53	0.8320	95.829
GUQINAGENT two-stage, w/o feedback	86.237	31.678	0.7616	18.339	14.99	0.7810	78.468
Our methods							
GUQINAGENT single-stage	98.732	**37.511**	0.7618	20.929	18.21	0.8113	14.258
GUQINAGENT two-stage	**99.981**	35.444	0.7301	17.013	**18.56**	0.7743	**0.194**
(b) Modern melodies: 10 scores, 91 work units							
Agent baselines							
GLM-5.3 ReAct	97.699	36.627	0.7118	39.961	9.73	**0.8543**	5.495
GLM-5.3 + GEPA	–	–	–	–	–	–	–
Direct supervision baseline							
Direct SFT	79.092	40.737	0.8364	43.437	19.12	0.8240	85.714
Ablations							
Fixed-workflow SFT, single-stage	19.298	9.409	0.7171	39.084	12.55	0.7199	96.703
GUQINAGENT two-stage, w/o feedback	73.485	38.099	**0.8623**	42.225	20.92	0.7643	85.714
Our methods							
GUQINAGENT single-stage	97.932	43.490	0.8358	**43.445**	**24.87**	0.7963	18.681
GUQINAGENT two-stage	**98.951**	**45.261**	0.8154	41.729	22.51	0.7774	**3.297**

Table 1: Main results on traditional scores (a) and modern melodies (b). Tech. Sim. and Tone Sim. range from 0 to 1. Other values are percentages.

Notes. [†]Original model: 11/59 attempted units valid; 972 skipped. [‡]Traditional Direct metrics use 1,003 valid contiguous-prefix units (97.28% coverage), not the full test; the rule-rate denominator is 1,003. N/A: unavailable; –: pending. Bold marks the best full-coverage result for metrics with a preferred direction. Metric definitions and conditional diagnostics are in the text and Appendix D.

Condition	Shared events	Pitch Acc. (%) ↑	Violation rate (%) ↓
Trajectory construction			
Fixed workflow	12,792	26.931	95.829
Expert-guided	12,792	98.781	14.258
Planning pipeline			
Single-stage	14,580	98.807	14.258
Two-stage	14,580	99.979	0.194
Diagnostic feedback in two-stage planning			
Feedback OFF	14,289	86.941	78.468
Feedback ON	14,289	99.979	0.194

Table 2: Traditional-score comparisons. Pitch accuracy uses the jointly resolvable events within each pair. Violation rates use all 1,031 work units per condition.

1,334 for two-stage. Consequently, the share of all targets that the auditor can verify as correct favors single-stage (89.856% vs. 86.387%). Reporting both measures distinguishes errors on resolved notes from differences in what the auditor can verify. The independence of these melodies from all development procedures remains to be verified.

Figure 3 compares pitch accuracy with diversity-adjusted nonbasic technique use and with estimated inference cost. On the modern set, the full system costs an estimated $0.824 per 100 attempted work units, compared with $1.770 for single-stage and $7.012 for GLM ReAct with the handbook. The two-stage feedback-off variant costs $0.370, showing the added workload associated with feedback and follow-up revision alongside the gains in shared-event accuracy. Direct SFT has the lowest estimated cost ($0.027) and lower pitch accuracy than either trajectory-trained variant. These estimates price retained visible requests and responses at public API rates without cache discounts. Appendix E specifies the accounting scope, and Ap-

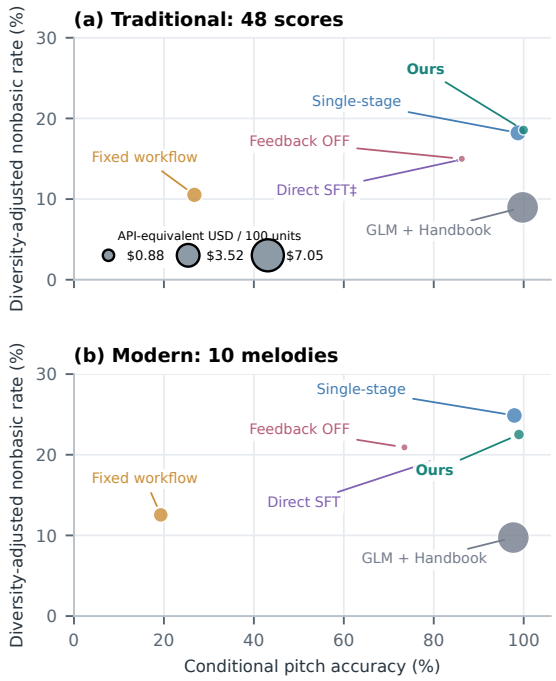


Figure 3: Pitch accuracy, diversity-adjusted nonbasic rate, and inference cost on traditional scores (top) and modern melodies (bottom). The vertical axis discounts nonbasic technique use by the probability of repeating a technique within the score (Appendix B). Bubble areas share a linear API-equivalent cost scale (USD per 100 attempted units; no cache discount). ‡Direct's traditional scores use 1,003 valid prefix units. Pitch accuracy is conditional on resolvability.

pendix F reports the remaining modern-melody diagnostics.

4.5 Expert Evaluation

We plan a blinded playing study on matched excerpts. Guqin performers will play anonymized outputs and rate two dimensions on a four-point scale: *performance feasibility* (practical string and fingering choices and continuity) and *idiomatic and overall musical expression* (appropriateness of techniques, phrasing, and audible *yunwei*). Appendix G describes the protocol.

[TODO: Report the evaluated excerpts, rater count, and ratings in Table 3. Summarize whether the two variants differ in feasibility or expression and how these judgments relate to the automatic metrics.]

5 Conclusion

GUQINAGENTframes jianzipu generation as executable, stateful plan editing. Its inverse trajectory construction turns private expert annotations into public tool-use supervision, supporting both single-stage generation and a two-stage pipeline for basic realization and idiomatic refinement. Both variants complete all 1,031 work units in the 48-score test. The full system achieves 99.981% pitch accuracy on auditor-resolvable events and leaves two runtime warnings, while our single-stage variant achieves higher reference agreement across fingering, tone type, and ornaments. On modern melodies, the full system also has higher conditional pitch accuracy, whereas single-stage generation resolves more target events and yields more verified-correct events overall. The fixed-workflow comparison shows that reproducing reference technique patterns alone does not ensure pitch consistency. Together, these results distinguish pitch fidelity, coverage, and reference agreement as complementary aspects of performance realization.

Method	Feasibility (1–4)	Expression (1–4)
GLM-5.3 ReAct	–	–
Direct SFT	–	–
GUQINAGENT single-stage	–	–
GUQINAGENT two-stage	–	–

Table 3: Planned expert evaluation. Method selection is provisional. Dashes indicate pending ratings.

6 Limitations, Ethics, and Resource Availability

The auditor is conservative and incomplete. Fine-grained left-hand reachability, timbre, and lineage-specific style require performer assessment or additional physical modeling. Inference feedback and final scoring share domain rules. Lower violation rates indicate closer adherence to the encoded playing practices. Independent performer assessment is needed to judge the resulting phrasing and expression. Generated notation also affects pitch resolvability; conditional accuracy must be read alongside coverage. The comparisons use different interaction budgets and trajectory-construction recipes, as specified in Section 4, and do not isolate every component's causal effect. The independence of the modern melodies from all development procedures remains to be verified. The digitized collection may not represent all historical sources or playing traditions. GUQINAGENTis intended to support performers and learners in preparing editable realizations. Performer evaluation remains pending (Appendix G), as does the GEPA compari-

son.

Open Science and Resource Availability. We will release the trained model, redacted training trajectories, the test benchmark, and the executable runtime used by teacher generation and inference. The release will include score-level split definitions, event-indexed schemas, evaluation scripts, filtering manifests, and instructions for reproducing the benchmark. Private teacher prompts, unredacted teacher reasoning, and any source material that cannot be redistributed will not be included. The archival repository URL, model license, dataset license, and versioned release identifier will be added before submission: [TODO: repository URL and licenses].

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A Training and Inference Configuration

Training. All four students use LLaMA-Factory with the `qwen3_5` chat template, bitsandbytes 4-bit quantization, BF16 computation, and LoRA targeting all supported linear projections. The saved adapters have rank 8, $\alpha = 16$, dropout 0, and no bias adaptation. Their target modules are `q/k/v/o_proj`, `gate/up/down_proj`, `in_proj_qkv`, `in_proj_z`, `in_proj_b`, `in_proj_a`, and `out_proj`. All use a cosine learning-rate schedule, warmup ratio 0.03, learning rate 2×10^{-4}, three epochs, and gradient checkpointing. Prompt tokens are not supervised (`train_on_prompt=false`); assistant history is not masked (`mask_history=false`). Direct targets contain only the final JSON and explicitly disable thinking. Its 3,976 cached targets are fully supervised without truncation; this is not a claim about the longer tool trajectories.

Setting	Two-stage	Single-stage recipes
Training scores	153	153
Underlying work units	3,976	3,976
Training conversations	7,952	3,976
Training GPUs	2	1
Batch per GPU	1	1
Gradient accumulation	6	12
Effective batch	12	12
Training cutoff (tokens)	8,192	8,192
Epochs	3	3
Optimizer updates	1,989	996

Table 4: Verified training configurations. Single-stage recipes comprise trajectory-trained single-stage, fixed-workflow SFT, and Direct SFT. The two-stage adapter is trained jointly on both stage types; OFF reuses that adapter without retraining.

Inference. Local Qwen inference uses vLLM with bitsandbytes 4-bit weights, the corresponding LoRA adapter, a 16,384-token context, and an 8,192-token per-request output limit. Tool-based adapted conditions use attempt temperatures $[0.6, 0.3, 0.0]$, top-$p = 0.95$, top-$k = 20$, min-$p = 0$, presence penalty 0, repetition penalty 1, and seed 20261024. A round is one model response, which may request tools; it is not necessarily one tool call. The limits apply per attempt, not to the aggregate trajectory. Two-stage ON permits up to three Guqinizer passes; OFF disables diagnostic follow-ups and uses one. Direct uses temperature 0, thinking disabled, no tool schema, and exactly one generation per attempted work unit. Within a score, all conditions preserve their own generated history rather than substituting reference plans. Failed predecessors cause subsequent work units to be skipped under the strict-history protocol.

GLM-5.3 ReAct + Handbook uses the production single-stage prompt, a fixed 73-entry handbook, eight rounds, two attempts, and temperature 0.25. Its saved launch configurations do not record the effective output-token cap or API context limit. The client code defaults to 8,000 output tokens but permits an environment override; we therefore do not present that default as a verified historical cap. GEPA remains pending and has no reported final evaluation budget or score.

Artifact fingerprints. Table 5 identifies the exact training JSONL files and final adapter weight files (`adapter_model.safetensors`) by full SHA256. The same checkpoint is used for traditional and modern evaluation within each method. Training configuration files in `train/configs/` use the suffixes `two_stage_final_2gpu_bs12`, `single_stage_final_1gpu_bs12`, `fixed_reference_ablation_1gpu_bs12`, and `direct_sft_1gpu_bs12`, each prefixed by `qwen35_9b_lora_` and ending in `.yaml`. Their independent data directories and preprocessing manifests bind the tokenized caches to the source data, template, and cutoff. The fixed-workflow prompts partly differ from the trajectory-trained single-stage prompts; this comparison is not a strict single-variable ablation.

B Evaluation Protocol and Baseline Details

Training scale and rollout limits. The training split contains 153 scores and 3,976 work units. Two-stage supervision yields 7,952 stage-specific conversations, jointly training one adapter used by both stages; single-stage trajectory supervision, fixed-workflow supervision, and Direct SFT each yield 3,976 examples. Candidate examples are screened for reference parsing and consistency, and generated traces are checked against the runtime before private expert content is removed from the student-facing sequences. All four students start from Qwen3.5-9B and use 4-bit LoRA fine-tuning for three epochs, with rank 8, scaling $\alpha = 16$, learning rate 2×10^{-4}, effective batch size 12, and an 8,192-token training cutoff. Equal epochs do not imply equal numbers of conversations or supervised tokens. Standard preprocessing truncation applies at the cutoff; counting a conversation does

Training recipe	Training JSONL SHA256	Adapter weights SHA256
Two-stage (ON / OFF)	6ea8329df61ade844389611d76ba5e32 15deba19f9a9aa6e7e0ae1d8c8d150b6 984779ea61a9c582328518bf8d4e0238	ecdee4141e31cabad373a7dba0fbcddc f0a03d15ff11f46bb41ca02ec510d0ad cdcc420e0b193cb1b6ad2520af13ae43
Single-stage	26bf31b8c705f927c1dfbd9d97b00f4a 72f664d51ca2aecb62e792d20043f8bf	b5549bac63cb8d4e3d7b0123daefb436 8447be00f62ad6a81c605b5d1cb59e7b
Fixed workflow	befe494c64d10c1a4b6f0e68c81ffc6a edba4abf12f1d2834bd079ea30f7ed1e	7149582b7caddec987e33ba9ee57ac29 30c0e8fcd8fb7387d86f66677f8596c2
Direct SFT	f50238b2a631db6948712b18d102f7e9	4327a693eced4f5ac8131dfcb27d0913

Table 5: Exact artifact identities. Concatenate the two lines in each hash cell without whitespace.

Component	Contents
Input	Complete-score metadata, tuning, jianpu events, and stable event indices.
Output	One event-indexed jianzipu plan for the complete score.
Runtime	Pitch candidates, editing, context retrieval, validation, and diagnostic feedback.
Reference	Expert notation for offline supervision and structured action comparison.

Table 6: Benchmark representation. Reference notation is withheld from inference.

not certify that every original token is supervised.

At inference, local Qwen conditions use a 16,384-token context and at most 8,192 output tokens per request. Tool-based single-stage conditions allow up to 16 rounds per attempt; two-stage conditions allow eight rounds per stage attempt, with at most three attempts and, when feedback is enabled, three Guqinizer passes. These are ceilings, not required numbers of interactions. Feedback OFF performs one Guqinizer pass. Direct generates once per work unit, without tools or repair. Thus these are deployed-pipeline comparisons, not equal-total-token experiments. Full configurations and artifact fingerprints appear in Appendix A.

Evaluation Metrics. Nonbasic Rate is the percentage of target-pitch events whose generated notation contains at least one nonbasic technique label and has no rule violation detected at that event. The implemented runtime rules identify violations through final-plan warnings. All target-pitch events remain in the denominator. It uses the ornament lexicon, excluding the eight isolated basic plucks and retaining recognized compound techniques; each event contributes at most once. The unfiltered reference usage rates are 31.65% for the traditional test and 40.38% for the modern test, or 31.88% on the traditional Direct prefixes.

Diversity-adjusted Nonbasic Rate. For score s, let N_s count evaluated target-pitch events and r_s be its Nonbasic Rate. Each qualifying event divides one count equally among its distinct nonbasic technique labels. Let $p_{s,t}$ be the share of these counts assigned to technique t. The adjusted rate is $A_s = r_s(1 - \sum_t p_{s,t}^2)$. The factor $1 - \sum_t p_{s,t}^2$ is the probability that two independent labels drawn from the qualifying usage distribution differ. Thus, the adjustment discounts the original rate by the chance of label repetition. For example, 20% usage split evenly across four techniques gives $20\% \times (1 - 1/4) = 15\%$. A score with zero or one qualifying technique label receives zero. We compute the rate directly from counts and report $100 \sum_s N_s A_s / \sum_s N_s$ in Table 1 and Figure 3.

Pitch correctness is measured against the input jianpu melody. The expert jianzipu annotation supplies reference actions for fingering and technique metrics. Let $\mathcal{I}$ contain events for which the auditor returns a determinate matched or mismatched prediction. Let $a_i = 1$ if event i matches the target within 50 cents under the supported matching rules, and $a_i = 0$ otherwise. Let $\mathcal{A}_i$ contain the tone pairs aligned by the auditor, with absolute error δ_{ij} in cents for pair j. We report

$$\begin{aligned} \mathrm{PitchAcc}_{50} &= \frac{\sum_{i \in \mathcal{I}} a_i}{|\mathcal{I}|}, \\ \mathrm{PitchMAE} &= \frac{\sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{A}_i} \delta_{ij}}{\sum_{i \in \mathcal{I}} |\mathcal{A}_i|}. \end{aligned} \tag{8}$$

Accuracy uses events as its denominator; MAE averages aligned tone pairs, and a compound event can contribute multiple pairs. Unresolved cases are excluded from pitch scoring and must be reported separately. We report each condition's evaluable event count and accuracy on their intersection, since generated notation affects which expressions the auditor resolves. The complete event inventory includes structural and continuation events and is not the pitch denominator.

Position F1 is a micro set F1 over exact $(\text{string}, \text{hui})$ pairs within each event, using the stateful auditor's resolved positions. All resolved positions of a compound event are included, without imposing a primary/secondary ordering. Open

strings receive an explicit OPEN contact label; numeric hui positions are normalized to six decimal places. Events with unresolved reference positions are excluded, whereas unresolved predictions at scorable reference events contribute missing pairs. The common reference coverage is 12,327/18,128 target events (68.000%) for traditional scores and 1,280/1,528 (83.770%) for modern melodies; the reference contains 12,821 and 1,480 scorable position pairs, respectively. This is conditional reference-position agreement, not a complete transcription score. Left-hand fingering, right-hand fingering, and tone type are reported in Appendix C. Tone type distinguishes open, stopped, and harmonic notes. Ornament precision, recall, and F1 compare event-level sets of named embellishment and compound actions. They share the technique-distribution lexicon and alias normalization below, but exclude the eight basic plucks (mo, tiao, gou, ti, da, zhai, tuo, and bo). Known compounds remain atomic labels. Harmonic-region start/end markers are retained only as state controls, never as techniques or ornaments. A nonempty disjoint label set scores zero F1; two empty corpus sets are undefined. These metrics measure agreement with one expert realization, not correctness of execution.

Technique distribution similarity summarizes the relative use of named performance techniques. Each technique is counted at most once per event, then counts are normalized over all technique incidences in the corpus. For predicted and reference distributions p and q over their joint technique vocabulary $\mathcal{V}$, we use

$$\mathrm{TechSim}(p, q) = 1 - \frac{1}{2} \sum_{t \in \mathcal{V}} |p(t) - q(t)|. \quad (9)$$

The score lies in $[0, 1]$, with higher values indicating closer distributions. This metric excludes notation-state markers such as harmonic-region boundaries. It does not evaluate event order, and identical distributions can assign techniques to different events; ornament F1 retains a local comparison. Tone-type distribution similarity uses the same formula over open, stopped, and harmonic notes and is reported in the appendix.

Rule Violation Rate measures the fraction of evaluated work units whose final plans contain at least one runtime-detected rule violation. We replay complete scores through the same rule checker, including for conditions that hide diagnostic feedback during inference; a final runtime warning is the operational indicator of a detected violation. Each work unit contributes at most one to the numerator, regardless of how many rules it violates. The denominators are 1,031 work units for the traditional test and 91 for the modern test in the complete-coverage comparisons. We separately report the total number of detected violations. Thus this metric measures final-plan rule compliance, not the frequency of warning messages emitted during inference. It covers only the implemented checks: unresolved expressions are not automatically violations, and absence of detected violations does not certify playability or musical quality.

Models and Comparisons. The feedback comparison uses Qwen3.5-9B with the same project-trained two-stage LoRA weights in both conditions. The full system exposes warnings, diagnostic hints, and warning-triggered follow-up interactions. The diagnostic-feedback-off condition hides this feedback and disables its follow-up mechanism while retaining structural validation and pitch candidates. Its public user and tool messages were checked for diagnostic-feedback leakage.

GUQINAGENT, single-stage is our trajectory-trained variant without the Fingering–Guqinizer decomposition. It uses a separately trained Qwen3.5-9B LoRA and generates final jianzipu through the same public tools and diagnostic feedback as the full system. It is allowed up to 16 rounds per attempt, while the full pipeline allows eight rounds per stage attempt and up to three Guqinizer passes. Retries and additional passes make their aggregate budgets different; this is a comparison of the two deployed pipelines. Both use complete model-generated histories and the same scoring rules. Table 1 marks both the single-stage and two-stage systems as Ours, and lists the two-stage feedback-off ablation above them.

Direct final-artifact supervision. We train a Qwen3.5-9B Direct SFT comparison using the same 3,976 training examples and final committed plans as our single-stage trajectory supervision. Each example retains the initial public musical context and one assistant JSON target, without intermediate reasoning, tool calls, or tool observations. Inference generates one final plan per work unit without tools or diagnostic revision, using the model's own successful preceding outputs as history. This comparison evaluates the end-to-end artifact-only workflow against trajectory-trained tool interaction; it changes both supervision and inference access to

tools. Direct completes all 91 modern work units, but its traditional-score output coverage is incomplete; conditional prefix diagnostics are reported separately in Appendix D.

We also evaluated the original Qwen3.5-9B checkpoint without the project LoRA using the two-stage agent and the same tool definitions. All 59 attempted inputs matched the test-input hashes. The original-model run allowed 8,192 output tokens per request, eight rounds per stage attempt, up to three attempts, and up to three Guqinizer passes, with a context limit of 16,384 tokens. After a work unit failed, the evaluator skipped the remainder of that score, preserving the requirement that history contain model-generated predecessors. These limits match the current SFT evaluation entry point.

The commercial-model ReAct + Handbook baseline uses GLM-5.3 through its API with the production single-stage prompt and a fixed 73-entry technique handbook. It allows eight rounds and two attempts at temperature 0.25, uses the same tools and scoring pipeline, and produces protocol-valid outputs for all 1,031 units. A planned GLM-5.3 ReAct + Handbook baseline with prompts optimized by GEPA on development data remains pending (Agrawal et al., 2026). These comparisons involve different model backends and do not isolate prompt optimization from model capability. GEPA's reflection model, API settings, optimization budget, and inference costs will be reported. Direct generation with the original Qwen checkpoint remains an optional additional comparison.

C Additional Action and Tone-Type Results

Table 7 reports event-level left- and right-hand fingering and tone-type F1, alongside tone-type distribution similarity. These complement the pitch, position, technique-distribution, ornament, and warning metrics in the main comparison. Tone similarity compares aggregate proportions of open, stopped, and harmonic notes and does not require these tone types to occur at the same events.

D Direct SFT Completion and Conditional Diagnostics

On traditional scores, Direct attempts 1,005/1,031 work units: 1,003 produce valid final JSON, two fail JSON parsing, and 26 are skipped after those failures. Valid-output coverage is 97.284%; attempted-unit protocol success is 1,003/1,005

Method	LH F1	RH F1	Tone F1	Tone Sim.
GLM + Handbook	43.609	40.835	54.555	**0.8439**
Fixed-workflow SFT	54.965	43.505	59.102	0.8320
GUQINAGENT, single-stage	**57.917**	**47.697**	**65.629**	0.8113
GUQINAGENT w/o feedback	56.599	45.659	61.670	0.7810
GUQINAGENT (Ours)	57.119	46.009	62.543	0.7743

Table 7: Additional structured action metrics on the same 48-score test set. F1 is a percentage; tone similarity is $1 - \mathrm{TVD}$ on $[0, 1]$. Higher values indicate closer agreement with reference annotations. All five conditions produce protocol-valid outputs for all 1,031 work units.

Method	LH F1	RH F1	Tone F1	Tone Sim.
GLM + Handbook	53.253	49.954	61.973	**0.8543**
Direct SFT	57.008	**59.361**	68.560	0.8240
Fixed-workflow SFT	52.970	46.019	65.576	0.7199
GUQINAGENT, single-stage	58.171	55.347	69.681	0.7963
GUQINAGENT w/o feedback	**59.350**	56.980	**70.049**	0.7643
GUQINAGENT (Ours)	57.255	59.010	67.663	0.7774

Table 8: Additional action metrics on ten modern melodies (91 work units), using source-score annotations as offline references. F1 is a percentage; tone similarity is on $[0, 1]$. All six methods have 100% protocol-valid coverage.

(99.801%), a different denominator. Both invalid responses end normally rather than reaching the output-token ceiling. We retain these failures and skips in completion reporting and do not regenerate them with reference or blank histories. Table 1 reports the available music metrics with an explicit conditional-prefix marker; they must not be interpreted as full-test estimates.

For diagnostic purposes only, we replay each valid contiguous prefix from the first work unit, preserving its generated history; we do not reset state at phrase boundaries. On these 1,003 valid units, conditional pitch accuracy is 85.277% on 14,909 resolvable events, with 57.291 cents MAE on 15,571 aligned tone pairs. Position F1 is 37.369, ornament F1 is 27.604, and technique similarity is 0.7536. Runtime replay detects 2,703 violations in 628/1,003 units (62.612%). These are prefix-conditional metrics, not full-test estimates. On 14,000 input events resolved by both Direct and the trajectory-trained single-stage student, pitch accuracy is 85.643% versus 98.821%; the latter is restricted to the same event intersection.

Modern Direct inference completes 91/91 work units. It attains 79.092% pitch accuracy on 1,387 resolvable events and 90.576 cents MAE, with 378 violations across 78 units (85.714%). On 1,315 events resolved by both Direct and trajectory-trained single-stage inference, pitch accuracy is 80.760% versus 98.023%. Direct ornament F1 (43.437) is close to the single-stage student's 43.445, despite substantially lower pitch consistency. This contrast supports distinguishing final-artifact agreement from pitch-grounded interactive realization, without attributing the entire gap to supervision alone.

The saved vLLM responses report 1,222,415 input and 218,369 output tokens for the 1,005 traditional attempts, and 114,109 input and 21,990 output tokens for the 91 modern attempts. Neither set contains an output-token truncation. There is one generation per attempted unit and no tool calls; traditional usage includes both invalid JSON responses. These reported usage counts are distinct from the retokenized API-equivalent estimates below and do not include skipped units as zero-cost model calls.

E API-Equivalent Inference Cost

Cost estimation. We distinguish recorded token usage from a reproducible API-equivalent cost estimate. For input tokens I, cache-hit input tokens H (a subset of I), and output tokens O, the token charge is

$$C = \left[r_{\text{in}}(I - H) + r_{\text{cache}}H + r_{\text{out}}O\right]/10^6, \quad (10)$$

where rates are USD per million tokens; cache creation and storage are additional when applicable. We use Together AI's public service prices checked on November 2, 2026: Qwen3.5-9B input/output rates are \$0.17/\$0.25, and GLM-5.3 input/cached-input/output rates are \$1.40/\$0.26/\$4.40.[1] Figure 3 uses model-specific public tokenizers and chat templates to retokenize the visible requests and responses retained with each final output. Qwen requests are saved verbatim; GLM inputs are reconstructed from saved user prompts and tool receipts with the production system prompt and fixed handbook. Server-side cache-hit counts were not retained, so we use an explicit no-cache-discount scenario ($H = 0$), not a claim of measured zero cache hits. These estimates exclude discarded runs, training, external tool execution, and unrecorded hidden reasoning or provider-only framing. Failed calls whose response usage was not saved are also excluded, not assigned zero cost; preflight context rejections are tracked separately. They describe retained visible-trajectory workload at public API rates, not complete historical usage, actual invoices, or local GPU expenses.

Method	Traditional	Modern
GLM + Handbook	7.046	7.012
Direct SFT	0.026	0.027
Fixed workflow	1.703	1.539
single-stage	1.829	1.770
feedback OFF	0.374	0.370
GuqinAgent (Ours)	0.700	0.824

Table 9: Estimated API-equivalent USD per 100 attempted work units, using the same visible-trajectory scope and no-cache-discount scenario in both datasets. Direct's traditional cost includes its two invalid responses and excludes skipped suffixes from the attempt denominator. These are retokenized estimates at public service rates, not historical invoices or self-hosting expenses. See Section E for the accounting definition.

F Supplementary Experimental Results

The traditional test contains 23,602 aligned event records across its 1,031 work units.

[1] https://www.together.ai/pricing; GLM rates also match https://docs.z.ai/guides/overview/pricing. These are provider list prices, not a manufacturer price for self-hosted adapters.

Original-model completion and protocol failures. The original-model agent attempted 59 of 1,031 work units across the 48 scores. Eleven attempts produced protocol-valid outputs (18.64% of attempted units), while 48 failed. The remaining 972 units were skipped because a predecessor failed. Valid-output coverage was therefore 11/1,031 (1.07%); skipped units are excluded from the attempted-unit protocol-failure denominator. All four tool-based adapted conditions produced valid outputs for all 1,031 units. We do not report full-test musical metrics for the original model or compare its 11 successful units directly with the adapted model's complete-test scores.

Of the 48 failed units, 13 failed during Fingering and 35 during Guqinizer. Logs record output truncation in 45 failed units, at least one attempt reaching round eight in 26, and three consecutive pitch-candidate-query rounds in 31. These observations overlap and are not a mutually exclusive attribution of failures. They suggest that sustained tool interaction and termination behavior are substantial obstacles for the original model under the current protocol, alongside the musical decisions required by the task.

Technique distributions and the commercial-model baseline. Technique distribution similarity is 0.7618 for single-stage SFT, 0.7616 for two-stage feedback OFF, and 0.7301 for the two-stage full system. Thus, fewer runtime warnings do not imply closer aggregate technique usage. GLM-5.3 with ReAct + Handbook attains 99.760% pitch accuracy and 2.542 cents MAE, with 102 final warnings across 52 units (5.044%). Its technique similarity is 0.6323 and ornament F1 is 7.158, below both trained students. Its tone-type distribution similarity is higher, however (Appendix C). These measures capture different aspects of realization; the cross-model comparison alone does not establish that trajectory training is necessary for idiomatic playing.

F.1 Traditional-Score Ablation Statistics

Pitch correctness and diagnostic feedback. The full system attains 99.981% pitch accuracy on 15,758 evaluable events, compared with 86.237% on 15,360 events when diagnostic feedback is disabled (Table 1). MAE is 2.308 versus 64.993 cents. Since the evaluable sets differ, Table 10 additionally reports accuracy on 14,289 events resolved in both conditions: 99.979% versus 86.941%, a gap of 13.038 percentage points. The observed accuracy gap therefore persists on the same input events. The result concerns auditor-supported realizations and does not establish accuracy on unresolved notation.

Diagnostic	ON	OFF
Evaluable pitch events	15,758	15,360
Aligned tone pairs for MAE	16,335	15,958
Common evaluable events	14,289	14,289
Pitch Acc. on common events (%)	99.979	86.941
Final runtime warnings	2	3,436
Work units with warnings	2	809
Rule violation rate (%)	0.194	78.468

Table 10: Diagnostic-feedback comparison. Counts refer to final plans, rescored under the same runtime with complete histories. Each condition covers 1,031 work units. OFF warnings are measured during evaluation despite being hidden during inference.

Residual warnings and error analysis. The full system leaves two final runtime warnings in two work units, compared with 3,436 warnings across 809 work units for OFF. The rule violation rate is 0.194% versus 78.468%. One remaining ON warning in *Youlan* is an octave error, realizing E6 for a target E7. These cases warrant inspection of inherited state, compound notation, and ordinary pitch choices. Final-warning counts and pitch-error counts use different eligibility and matching rules and should not be treated as interchangeable metrics.

Fingering agreement and idiomatic realization. The full system's position, left-hand, right-hand, and tone-type F1 scores are 35.444, 57.119, 46.009, and 62.543, respectively. Their modest differences from OFF accompany a much larger pitch-accuracy gap. The full system also has lower ornament F1: 17.013 versus 18.339. The current comparison provides evidence of improved pitch consistency without establishing an advantage in ornament selection.

Tone distributions indicate that pitch accuracy leaves room for different realization strategies. Among recognized tone types on reference-labeled events, harmonics account for 7.51% of ON predictions, compared with 29.34% of reference tone types. A system can realize the correct pitch through different strings, positions, or tone types, and can also underuse stylistically appropriate techniques. Both possibilities motivate examining action distributions and consulting performers when assessing idiomatic quality.

Diagnostic	Single-stage	Two-stage
Evaluable pitch events	15,935	15,758
Common evaluable events	14,580	14,580
Common-event Pitch Acc. (%)	98.807	99.979
Final runtime warnings	381	2
Work units with warnings	147	2
Rule violation rate (%)	14.258	0.194

Table 11: Single-stage and two-stage full-system comparison on the same 48 scores, scored after replaying complete histories. Both achieve 100% valid-output coverage. Different retry and pass structures preclude interpreting this as a strictly equal-total-budget architecture ablation.

Diagnostic	Fixed	Teacher
Evaluable pitch events	13,556	15,935
Common evaluable events	12,792	12,792
Common-event Pitch Acc. (%)	26.931	98.781
Final runtime warnings	11,596	381
Work units with warnings	988	147
Rule violation rate (%)	95.829	14.258

Table 12: Single-stage trajectory-construction comparison. Fixed uses a scripted query-and-edit sequence; Teacher uses expert-guided interactive trajectories. Both produce valid outputs for all 1,031 units. Some training prompts also differ between the recipes.

Our single-stage and two-stage variants. The single-stage student completes all 1,031 work units, with 98.732% pitch accuracy on 15,935 evaluable events and 7.880 cents MAE. The two-stage full system reaches 99.981% on 15,758 events and 2.308 cents MAE. On their 14,580 common evaluable events, accuracy is 98.807% for single-stage and 99.979% for two-stage, a gap of approximately 1.173 percentage points. Table 11 reports denominators and final warnings for this comparison.

The single-stage student has higher reference agreement in every reported action category: Position F1 is 37.511 versus 35.444, left-hand F1 is 57.917 versus 57.119, right-hand F1 is 47.697 versus 46.009, tone-type F1 is 65.629 versus 62.543, and ornament F1 is 20.929 versus 17.013 (Table 1). The two-stage pipeline has fewer final warnings, 2 versus 381, with rule violation rates of 0.194% versus 14.258%. These results show a trade-off: the current two-stage pipeline achieves greater pitch and runtime-rule consistency, while single-stage generation more closely reproduces the reference actions and ornaments. Performer assessment is needed to determine how these differences affect execution and musical expression.

Teacher trajectory construction. We compare our single-stage variant with a student trained on fixed-workflow trajectories. Each fixed trajectory contains generic reasoning, one real pitch-candidate query, and one edit constructed from the original expert annotation, without a teacher API rollout. Our teacher trajectories instead supervise the teacher's final committed plans after interaction. Both students are evaluated with diagnostic feedback and a 16-round stage limit on the same 48 scores using complete generated histories. The recipes differ in interaction traces, final targets, and some system/user prompts; this comparison evaluates the complete construction recipes rather than isolating reasoning supervision.

The fixed-workflow student achieves 100% protocol-valid coverage, but pitch accuracy is 26.778% on 13,556 evaluable events and MAE is 517.624 cents. The standard single-stage student attains 98.732% on 15,935 events and 7.880 cents MAE. On their 12,792 common evaluable events, accuracy is 26.931% versus 98.781% (Table 12). Thus, the large pitch gap persists on a shared event set rather than arising solely from different parseability.

The fixed-workflow model leaves 11,596 warnings across 988 units, including 9,926 pitch-mismatch warnings. Its Position F1 is 11.002 versus 37.511 for the standard single-stage student. However, its ornament F1 is higher (27.928 versus 20.929), as is technique distribution similarity (0.8401 versus 0.7618). Matching reference technique usage therefore does not ensure that those actions realize the input pitches. The results favor expert-guided trajectories for pitch consistency under the evaluated recipes, while leaving the separate roles of reasoning content, iterative feedback, prompt differences, and training length unresolved.

F.2 Modern-Melody Diagnostics

We evaluate the two-stage system, single-stage student, Direct SFT, fixed-workflow student, diagnostic-feedback-off variant, and GLM-5.3 ReAct + Handbook on the same ten modern melodies (91 work units) using complete generated score histories (Table 1, lower panel). All six methods produce protocol-valid outputs for every unit. Pitch is audited against the input melody; source-score jianzipu annotations provide reference actions for agreement metrics. The sealed references match all 91 public input IDs, hashes, and runtime inputs,

and are used only for offline evaluation. These annotations represent one realization, not an assessment by the performers in our planned study. The independence of these melodies from all training and model-selection procedures remains to be verified, so this analysis is not yet a strictly held-out generalization benchmark.

Figure 3 compares conditional pitch accuracy, diversity-adjusted nonbasic rate, and estimated inference cost across methods. Direct uses substantially fewer tokens but has lower conditional pitch accuracy. Its traditional point is prefix-conditional, not a full-test estimate. Bubble areas use the same retokenized visible-trajectory estimator for every method. Direct's two invalid traditional responses contribute to workload; cost is normalized by 1,005 actual attempts rather than assigning zero-cost calls to 26 skipped units. Its separately recorded server usage remains in Appendix D.

Of 1,528 events with a nonempty target pitch, 1,334 are auditor-resolvable for the two-stage system and 1,402 for the single-stage system, corresponding to 87.304% and 91.754% coverage. Conditional PitchAcc@50c is 98.951% and 97.932%, respectively; the proportion of all 1,528 target events both resolved and correct is 86.387% and 89.856%. On the 1,248 events resolvable for both, the corresponding accuracies are 99.920% and 98.157%. Full-score runtime replay yields 15 warnings across three work units for two-stage, versus 42 across 17 for single-stage. These are rule-based diagnostics, not independent performer judgments; the stage and retry budgets also differ between methods.

ReAct + Handbook attains 97.699% pitch accuracy on 1,347 auditor-resolvable events, with 16.035 cents MAE and 33 warnings across five units. Its Position F1, technique similarity, and ornament F1 are 36.627, 0.7118, and 39.961, respectively. As on traditional scores, the single-stage student's reference agreement can exceed that of the two-stage system despite lower conditional pitch accuracy. Disabling diagnostic feedback yields 73.485% pitch accuracy on 1,403 evaluable events and 472 final warnings across 78 units. On the 1,271 events resolvable for both ON and OFF, accuracy is 98.899% versus 73.879%. OFF's technique similarity (0.8623) exceeds the full system's, again showing that aggregate reference usage and pitch consistency capture different properties.

The fixed-workflow student obtains 19.298% pitch accuracy on 1,197 evaluable events, with 558.105 cents MAE and 1,060 warnings across 88 units. On the 1,119 events resolvable for both single-stage students, the teacher-trajectory student attains 98.749% versus 19.929% for fixed workflow. Thus, the large construction-recipe gap also occurs on modern melodies and persists on shared events. Prompt and training-trajectory differences still limit causal attribution to any individual construction step.

Direct SFT completes all modern inputs but attains only 79.092% pitch accuracy and an 85.714% rule violation rate, despite ornament F1 (43.437) close to the trajectory-trained single-stage model (43.445). On 1,315 jointly resolvable events, Direct and trajectory-trained single-stage pitch accuracies are 80.760% and 98.023%, respectively. Final-artifact agreement therefore does not imply reliable pitch realization. This pipeline comparison changes tool access and inference budget as well as supervision, so it does not isolate their individual effects.

G Planned Expert Evaluation

A blinded performer study is planned with one or two guqin experts who will actually play selected scores. Each will rate two dimensions on a four-point scale: *performance feasibility*, covering string and fingering choices and continuity, and *idiomatic and overall musical expression*, covering the suitability of techniques and audible *yunwei*. Scale anchors and sampling will be fixed before assessment. Examples will be anonymized and presented in randomized order, with input scores available to raters. We will report rater experience and sampling, and inter-rater agreement if two experts assess the same examples. Expert scores and claims of superior playability or expression await this assessment.